

# Essentials of Blockchain

# Types of Blockchain Platform

- Public - Permissionless (anyone j/r/w)
- Private - Permissioned - (organization)
- Consortium- Permissioned(Flexible)

# Public

- Anyone can join ( read/write/participate)
- Decentralized platform - no contro of anyone
- Open for all - no rules
- user are anonymous - security - they do not know each other
- Transperent
- Slow
- eg: Bitcoin

# Private

- Permissioned
  - Single organization - there is doubt in data
  - Partially decentralized
  - Abide rules - who can read , who can write
  - Low transaction fees - (lesser n/w, lesser no. of participants are there)
- 
- it is called an enterprize blockchain
  - eg: hyper ledger

| Feature           | Public                             | Private                                       |
|-------------------|------------------------------------|-----------------------------------------------|
| Access            | Permissionless                     | Permissioned                                  |
| Authority         | Decentralized                      | Partially Centralized                         |
| Transaction Speed | slow                               | Fast                                          |
| Consenses         | Permissionless                     | Permissioned                                  |
| Transaction Cost  | High                               | Low                                           |
| DAta Handling     | Read and Write access for everyone | Read and write access for single organization |
| Immutability      | Ful                                | Partial                                       |
| Efficiency        | Low                                | High                                          |

# Open or Close

- Who can read
- Open-anyone can read
- Closed-Restricted read
- We have-
  - Public open - permission less - **who can write**- open for write and open for reading
  - Public close - - open for write but closed for reading
  - Private open - premissioned .....writing is pri and reading is open
  - Private close- write private read - restricted
- these permissions are given by identity access mgmt ---- Google,ibm,fb, or its own - for (which user can access what)
- ethereum - public platform - anyone can use
- permission - user

# Eg : Diff. Applications

## Public and Closed

- Voting
- Voting Records

## Public Open

- Currencies
- Betting
- Vedio Games

## Private Closed

- Construction
- National Defence
- Military
- Tax Returns

## Cont...Depends on application

### Private Open

- Supply Chain (Vendors can write only..others can read only)
- Govt Financial records
- Corporate Earning Statements



# Consortium Blockchain Platform

- Multiple Organization Decentralised
- ....Similar to private chain platform, only difference is in private chain platform,,Single organization is dealing and in consortium...multiple organizations collaborate together, so it becomes fully decentralized as more than one organization is controlling it.
- And the later is partially decentralized.

Permissionless

Permissioned

**Public**

No central Authority

**Hybrid**

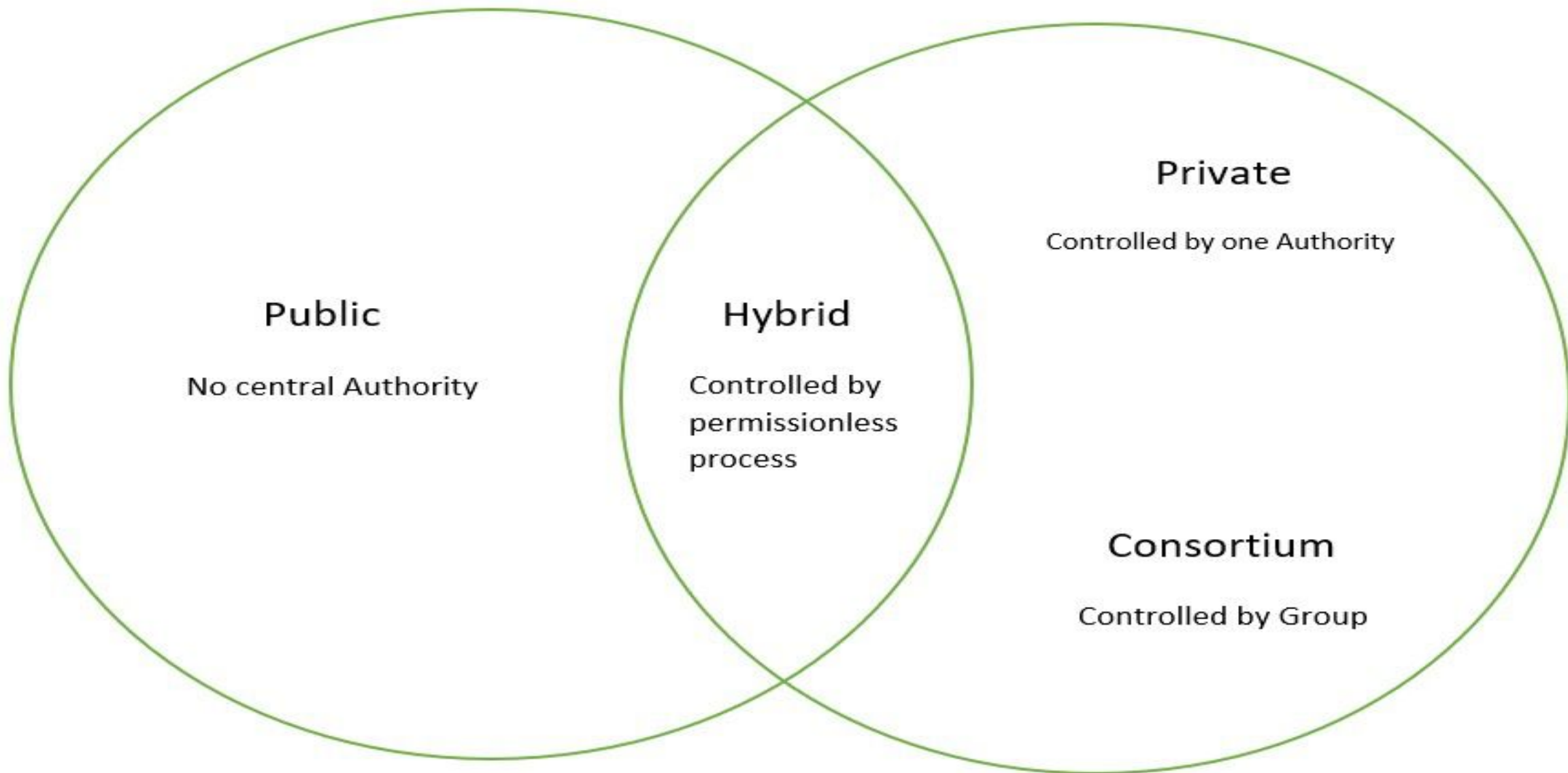
Controlled by  
permissionless  
process

**Private**

Controlled by one Authority

**Consortium**

Controlled by Group



How does it Work????